

TASK 23.09.2024

1. For each of the entities listed below, find the attributes and the UID.

- a) A teacher
- b) A customer
- c) A student
- d) A book
- e) A transaction
- f) A car

2. Identify the possible Entities and their Attributes from the given scenarios. Write all the entities and for each of them, write a list of attributes

S1. In a computer company there are several departments, each one being identified by a code. For each department the name of the department is known, as well as the manager. Each department has several offices, located in one of the company's buildings. An office may have many printers but there may not be any printers in some offices. The company keeps track of companies that offer toners and cartridges for each type of printer. It is known for each such company at what price each type of toner or cartridge is provided. A company may not have toners or cartridges for all types of printers the company owns. For each company, the order honoring period is known.

S2. A Country Bus Company owns a number of buses. Each bus is allocated to a particular route, although some routes may have several buses. Each route passes through a number of towns. One or more drivers are allocated to each stage of a route, which corresponds to a journey through some or all of the towns on a route. Some of the towns have a garage where buses are kept and each of the buses are identified by the registration number and can carry different numbers of passengers, since the vehicles vary in size and can be single or double-decked. Each route is identified by a route number and information is available on the average number of passengers carried per day for each route. Drivers have an employee number, name, address, and sometimes a telephone number.

S3. "We have several teachers at our school. A teacher can be assigned up to three classes per semester. If a teacher is on sabbatical, he doesn't teach that semester. We keep a record of the teacher's first name, last name, address, phone number, and email address. Our school offers many courses -- such as Data Modeling, Introduction to SQL, Trigonometry, Physics, and Biology. Each course has a code. For example: Data Modeling would be DM001, Trigonometry would be TR004, etc. During each semester, a course may be taught in several classes - so there could be two classes of Physics, three classes of Biology, etc. Each class can be taught by only one teacher. We assign a unique ID for each class, and we also keep track of the day it is taught, the time, and the classroom."

S4. You must manage the activity of a pharmacy. The pharmacy sells several drugs and has a price for each drug. Patients who buy drugs are identified by an ID, age and address. For each drug the name is known, which is unique and the composition. Each drug is distributed by a pharmaceutical company that may have several contracts with the pharmacy. For each contract, we must know the date of signing the contract, the date on which it expires and a description of it. For each pharmaceutical company, the name, address, telephone number and name of a contact person within the company are known. Doctors who issue prescriptions issued by the pharmacy are identified by an ID, but the name, speciality, number of years of experience are memorized. A doctor may prescribe several medicines to a patient, and a patient may receive treatment from several doctors. Each prescription is listed on the date it was issued and contains a list of prescribed drugs, the required amount, the doctor's name and ID, the patient's name and ID.